

Building of NETFLIX using System Design

NETFLIX

Functional

Requirements

- * Users can register & login
- * User can buy a Subscription
- * User can play or pause a video

Non-functional Requirements

- * Secure system with authentication & authorization
- * Low latency for videos
- * Scalable Systems

→ LLD ~~is~~ Completed. Goal:-

→ HLD:-

1. Fundamentals
2. Database
3. Consistency vs Availability
4. Cache
5. Networking
6. Load Balancing
7. Message Queues

8. Monoliths vs Microservices
9. Monitoring & Logging
10. Security
11. System Design Tradeoffs.
12. Practice:-

1. Youtube
2. Twitter
3. Whatsapp
4. Uber
5. Amazon
6. Dropbox / Google Drive
7. Netflix

8. Zoom

9. Instagram

10. Booking.com / Airbnb

→ HLD :- [High level design]:

Building Blocks

Design goals

- Scalability
- Availability
- Single point of failure
- Consistent
- Eventually Consistent

of System Design

Network

- IP Address
- DNS
- Cloud Computing
- Client & Server
- Protocols
- TCP
- UDP
- HTTP
- Websockets
- Forward Proxy
- Reverse Proxy

Database & Storage

- SQL
- No SQL
- SQL vs NoSQL
- Object Storage
- CDN
- Cache

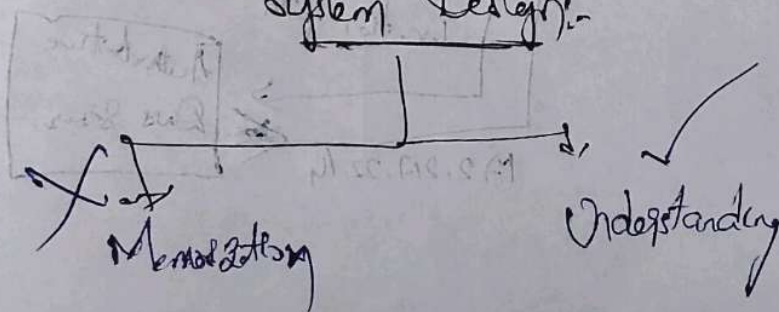
Communication

- API
- GraphQL
- gRPC
- Message Queues
- REST API

⇒ Extras -

- Cache Aside strategy
- Read Through Cache
- Logging & Monitoring

System Design:-



→ IP Address:-

Every each Computer has a Unique IP Address

Ex:- 172.217.22.14

→ Computer introduces themselves to each other on the Internet.

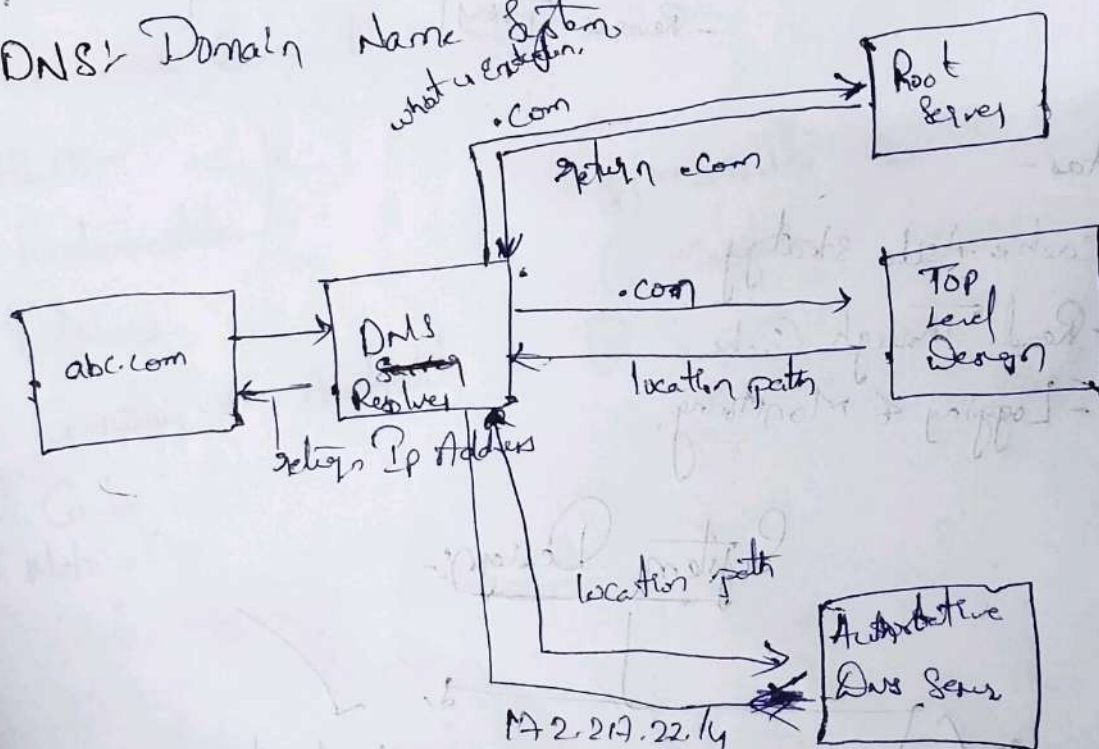
Types of IP Address:-

→ IPv4 :- [172.217.22.14]

→ IPv6 :- [2001:db8:3333:4444:5555:6666:7777:8888]

→ Every IP Address is Connected to the Internet.

→ DNS: Domain Name System



→ Cloud Computing: -

* Instead of using physical machine may cause

* time consuming

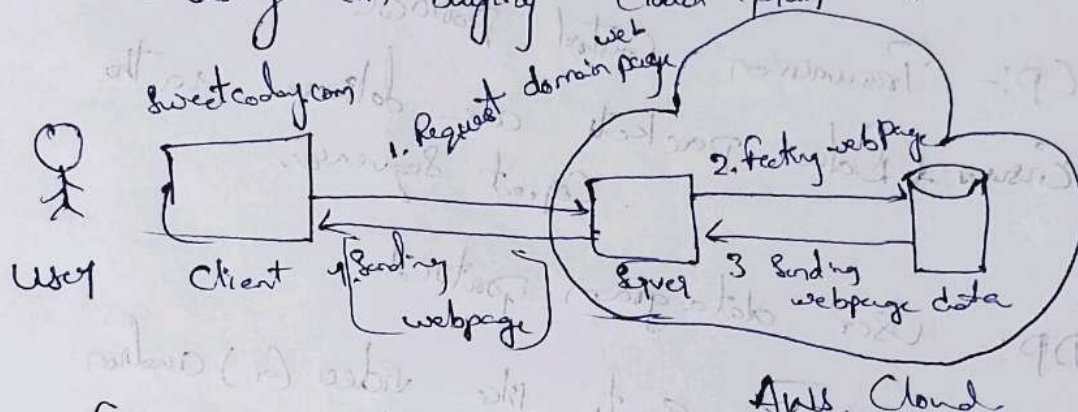
* Expensive

* replacements when physical machine broke.

* maintenance

* power consuming

* Using (a) buying cloud platforms.



* Benefits of Cloud Computing: -

1. Scalability

2. Super Cost effective.

3. Reliability

→ monitor

→ Backup

→ Security.

→ Client & Server

* request information from the server. (mobile, pc)

Server

* A computer that from the client

Server

the requested information

TV → netflix

* Netflix Info.

* Protocols:-

* Based on the task we use different protocols.

Ex:- Common web Interaction (web pages)
* HTTP

Ex:- Transferring of files-

* FTP (file transfer protocol),

→ TCP:- Transmission Control protocol

Ensures Data packets are delivered in the
Correct Sequence.

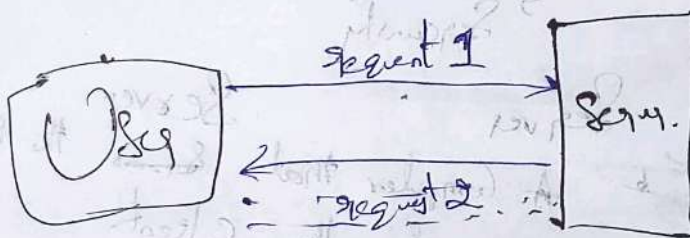
→ UDP:- User datagram protocol.

* Live Events like video (s) and/or

* It does use datagrams

* It is fast.

→ WEBSOCKETS:-



Bidirectional Conversation